ABSTRACT

of the PhD thesis by Yesmukhamedov Nurmaganbet Seitkaliuly on «Automated Diabetic Retinopathy Diagnosis via Fundus Image Enhancement and CNN Classification», submitted for the degree of Doctor of Philosophy (PhD) in the educational programme (EP) 8D06104 — Computer systems and software engineering

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General characteristics of the research

This dissertation investigates and formalises the integration of fundus image enhancement (preprocessing) with convolutional-neural-network (CNN) classification for automated multi-stage diabetic retinopathy (DR) diagnosis. Its central idea is that preprocessing is an integral component of the diagnostic model rather than ancillary data preparation, because it defines the feature space available to the network. To operationalise this idea, an 8-stage preprocessing pipeline is specified and embedded in the model, and its effect is placed under a controlled experimental contrast against an equivalent baseline trained without the pipeline. The approach is validated across eight public and clinical fundus datasets acquired with cameras of four manufacturers, covering diagnostic dominance, component ablation, direct measurement of domain-shift reduction, cross-dataset transfer, explainability, external clinical performance, and device domain shift, and is oriented towards automated DR screening in resource-limited settings, including the healthcare infrastructure of Kazakhstan.

Relevance of the research topic

Diabetic retinopathy (DR) is one of the leading causes of preventable blindness among the working-age population worldwide. Because the early stages of DR are clinically asymptomatic, timely detection depends on regular screening of fundus images, yet the number of ophthalmologists is insufficient to manually examine the rapidly growing population of diabetic patients — a shortage that is especially acute in resource-limited and rural healthcare settings, including the regions of Kazakhstan. Automated diagnosis based on convolutional neural networks (CNNs) is therefore a practically significant direction of research.

The dominant tradition in automated DR diagnosis — here designated paradigm **P1**, the end-to-end CNN paradigm — treats image preprocessing as ancillary data preparation that does not require methodological discussion, on the assumption that a sufficiently deep network learns the relevant invariances directly from raw or minimally normalised pixels. In practice, however, fundus images acquired by different cameras, under different illumination conditions, and with different noise levels exhibit substantial domain variability. This variability manifests as distribution shift in the input feature space and degrades the generalisation of CNN models trained on one domain when they are applied to another. The methodological gap motivating this dissertation is the absence of a formalised, experimentally controlled treatment of preprocessing as an integral component of the diagnostic model.

This work advances an alternative paradigm — **P2**, the integrated preprocessing-CNN paradigm — in which preprocessing is an integral component of the model itself, because it defines the feature space available to the network and therefore co-determines what the network can learn. The relevance of the topic follows from the need to reduce inter-device and inter-acquisition domain variability while preserving diagnostically relevant retinal features, and to do so under the computational constraints characteristic of real screening environments.

Aim of the research

To develop and experimentally validate an integrated fundus image enhancement and CNN-based classification framework for automated multi-stage diabetic retinopathy diagnosis, in which an 8-stage preprocessing pipeline is treated as an integral component of the model, producing a statistically measurable and reproducible improvement in five-class DR diagnostic performance relative to an equivalent architecture trained without the pipeline, under constrained computational conditions.

Objectives of the research

1. To analyse the medical, epidemiological, and technical context of automated DR diagnosis, the sources of fundus image quality variability, and the current state of CNN-based screening systems, and to formulate the research problem (Chapter 1).
2. To establish the theoretical foundations of image enhancement (histogram equalisation, dual-constraint CLAHE, spatial filtering), of convolutional neural networks (convolution, pooling, loss functions for imbalanced data, regularisation), of transfer and self-supervised representation learning, and of explainability methods (Grad-CAM, ALO, IoU) (Chapter 2).
3. To design and formalise the integrated preprocessing-CNN methodology — the 8-stage pipeline, the CNN architectures (ResNet-50, EfficientNet-B3), the pretraining strategy, and the multi-metric evaluation framework (Chapter 3).
4. To experimentally validate the integrated pipeline through eight investigations conducted on a common substrate: integrated-pipeline dominance (Experiment 1, H-1); component ablation with the CLAHE and flat-field parameter sweeps (Experiment 2, H-2); direct measurement of source-to-target distance in feature space across six external corpora (H-3); cross-dataset transferability (Experiment 3, H-4); Grad-CAM explainability (Experiment 4, H-5); external clinical performance (Experiment 5, H-7); device domain shift (Experiment 6, H-6); and small-data training (Experiment 7) (Chapter 4).
5. To validate the reliability of the results through statistical analysis and comparative benchmarking, and to state the limitations and boundary conditions of the approach (Chapter 5).

6. To design an architecture for an automated DR screening system suitable for resource-limited environments, with telemedicine and eHealth integration applicable to Kazakhstan healthcare infrastructure (Chapter 6).

Object of the research

The process of automated diagnosis of diabetic retinopathy from colour fundus images using convolutional neural networks.

Subject of the research

Methods and models for integrating fundus image preprocessing (enhancement) with CNN classification, and the effect of the integrated preprocessing-CNN pipeline on the diagnostic performance, transferability, interpretability, and device robustness of five-class DR classification.

Methodology and methods of the research

The research methodology combines image-processing theory, deep-learning theory, and an experimentally controlled validation framework. The methods used include: classical computer-vision detection (optic disc and fovea localisation for rotation normalisation), adaptive flat-field illumination correction, dual-constraint Contrast-Limited Adaptive Histogram Equalisation (CLAHE) in the LAB colour space, CNN classification with ResNet-50 and EfficientNet-B3 backbones, transfer learning from ImageNet for the baseline arm and in-domain pretraining for the integrated arm — the choice among candidate initialisations being settled by a frozen-backbone linear-probe acceptance gate rather than assumed — Focal Loss for class imbalance, Grad-CAM explainability analysis with Attention–Lesion Overlap (primary) and IoU (secondary), maximum mean discrepancy over penultimate-layer features for the direct measurement of domain shift (with per-channel histogram divergence as informational context), and a statistical validation framework based on 5-fold patient-level stratified cross-validation, McNemar and DeLong tests, mixed-effects modelling, bootstrap confidence intervals, and Bonferroni/Holm correction for multiple comparisons.

Empirical (experimental) basis

Eight datasets organised into functional tiers: EyePACS (~35,126 images, primary training and ablation, Canon CR-1); APTOS 2019 (~3,662, cross-dataset transfer); IDRiD (516 images, 81 with pixel-level lesion masks, clinical validation and explainability, Kowa); Messidor-2 (~1,748, external clinical evaluation, Topcon); DDR (~13,673, device shift, Canon/Topcon); ODIR-5K (~5,000, device shift, Canon/Zeiss); RFMiD (~3,200, device shift, Topcon/Kowa); and a Kazakh clinical dataset (60 images, 30 patients × 2 eyes, balanced, qualitative validation). Image quality is evaluated with contrast-to-noise ratio (CNR), image entropy, and SSIM.

Scientific novelty

1. An 8-stage integrated fundus preprocessing pipeline is formalised as a binding component of the diagnostic model (operationalisation of paradigm P2), comprising canonical flip, OD-fovea rotation normalisation, FOV crop with isotropic resize and centred zero-padding, FOV mask generation as a 4th input channel, adaptive flat-field correction ($\sigma = 0.07 \times$ FOV diameter), dual-constraint stochastic CLAHE on the LAB L-channel, augmentation, and dataset-specific channel-wise normalisation.
2. The hypothesis of integrated-pipeline dominance is formalised and tested under a controlled 2×2 factorial design (configurations A–D) over two established architectures (ResNet-50, EfficientNet-B3), with a pre-registered dominance criterion (Δ weighted F1 $\geq$ 5 pp, Δ ROC-AUC $\geq$ 0.02, no Cohen's Kappa degradation).
3. A dual-constraint stochastic CLAHE variant is adapted and validated within the five-class DR context (LAB L-channel; clip limit CL = min(clip_factor × tile_area / 256, global_threshold × tile_area); 80% train-time probability).
4. An adaptive flat-field illumination correction is introduced whose Gaussian σ scales with the per-image FOV diameter rather than using a fixed global value, applied only inside the FOV mask.
5. An explicit binary FOV mask is introduced as a dedicated pipeline stage and appended as the 4th input channel, informing the CNN of valid pixel regions and preventing padding artefacts from being learned as features.
6. Attention–Lesion Overlap (ALO) is introduced as a primary, asymmetric explainability metric that directly measures the fraction of a lesion covered by model attention, complemented by Intersection-over-Union (IoU) as a secondary metric, evaluated against IDRiD pixel-level lesion masks.
7. The mechanism postulated by the central hypothesis is measured directly rather than inferred from its consequence: source-to-target distance is computed at the penultimate layer under both configurations across six external corpora, at the cost of forward passes only. Two design features carry the contribution — normalisation uses source-domain statistics and is never recomputed on the target, so the measurement is a property of the preprocessing and not of a procedure fitted to the target; and its falsifiability was recorded before it was made, since two stages of the pipeline are bound to the source domain by construction and a reversal was therefore a live possibility.
8. The integrated arm is initialised from in-domain rather than natural-image pretraining, with the choice among candidate initialisations settled by a frozen-backbone linear-probe acceptance gate rather than by assumption. The negative outcome is reported as evidence: label-free self-supervision trained from scratch on the in-domain corpus failed that gate, across several protocols of the same family and without improvement from longer training, and was not admitted.
9. The integrated pipeline is validated across a multi-dataset, multi-device architecture (EyePACS, APTOS 2019, IDRiD, Messidor-2, DDR, ODIR-5K,

RFMiD, and a Kazakh clinical dataset), including cross-dataset transfer, external clinical performance, and device domain shift across four camera manufacturers (Canon, Topcon, Kowa, Zeiss).

10. A defect shared by three measures in common use for expressing external robustness is identified analytically: the degradation quantity, the generalisation ratio and the retention ratio each normalise or difference an arm's external performance against that same arm's in-domain performance, and therefore penalise a configuration for its in-domain strength. The analysis is strictly descriptive and rehabilitates no result of this work.

Main results of the research

1. A critical analysis of the medical, epidemiological, and technical context of automated DR diagnosis and of fundus image quality variability was carried out; the limitations of the end-to-end paradigm (P1) were identified and the research problem was formulated.
2. An 8-stage integrated preprocessing pipeline was designed and formalised as an integral component of the diagnostic model (paradigm P2), comprising canonical flip, OD-fovea rotation normalisation, FOV crop with isotropic resize and centred zero-padding, FOV mask generation as a 4th input channel, adaptive flat-field correction, dual-constraint stochastic CLAHE, augmentation, and dataset-specific normalisation.
3. The dominance of the integrated configuration over the baseline was established under a controlled 2×2 factorial design (configurations A–D) on EyePACS for both ResNet-50 and EfficientNet-B3, evaluated against the pre-registered dominance criterion (Experiment 1).
4. The contribution of the individual pipeline stages was quantified through a cumulative component ablation conducted under a single initialisation, and the dual-constraint CLAHE clip-limit and the flat-field σ were characterised by parameter sweeps supported by image-quality metrics (CNR, entropy, SSIM) (Experiment 2).
5. The reduction of source-to-target distance in feature space was measured directly across six external corpora under a mandatory source-domain-statistics condition, supplying the middle term of the causal chain that the remaining hypotheses approach only through its consequence (H-3).
6. The cross-dataset transferability of the pipeline was evaluated on APTOS 2019 without retraining against the pre-registered generalisation ratio $G \geq 0.85$ (Experiment 3).
7. Grad-CAM explainability analysis, using the proposed Attention–Lesion Overlap (ALO, primary) and IoU (secondary) against IDRiD pixel-level lesion masks, characterised the alignment of model attention with clinically annotated lesion structures (Experiment 4).
8. The external clinical performance of the integrated configuration (IDRiD, Messidor-2; Experiment 5), its device robustness across four camera manufacturers (DDR, ODIR-5K, RFMiD; Experiment 6), and its trainability on small clinical data (Experiment 7) were evaluated.

9. A modular architecture for an automated DR screening system suitable for resource-limited environments, with telemedicine and eHealth integration applicable to Kazakhstan healthcare infrastructure, was designed as a design specification; no prototype was implemented and no field testing was conducted.

Statements submitted for defense

Each statement is submitted at the strength its pre-specified criterion supports, and each carries a qualification that is not detachable from it: a statement given without its qualification is a different and stronger proposition than the evidence supports.

1. Preprocessing of fundus images is a formalisable and experimentally testable component of the diagnostic model rather than ancillary preparation of the data — the reframing from the end-to-end paradigm (P1) to the integrated paradigm (P2). *This statement is methodological and non-empirical: the experimental results are consistent with it under the conditions tested and do not establish it universally.*
2. The integrated configuration dominates the baseline configuration in five-class DR classification on EyePACS, on both ResNet-50 and EfficientNet-B3, under the conjunctive pre-registered criterion in all three of its components, with the effect surviving correction for multiple comparisons and showing no interaction with the choice of architecture (H-1). *The two arms differ in initialisation as well as in preprocessing, so the statement concerns the configuration as a whole; the cumulative ablation decomposes that composite but does not dissolve it, and no part of the effect is attributed to preprocessing alone.*
3. The dual-constraint CLAHE parameters and the flat-field σ each exhibit an interior optimum within the ranges examined, confirmed on held-out data (H-2). *Bounded to the ranges tested; the optimal values are properties of this corpus and pipeline rather than portable constants.*
4. The contributions of the individual pipeline stages are separable, monotone across folds, and orderable with the two photometric stages leading (H-2). *At the resolution of groupings only: adjacent ranks lie within noise, no strict ordering of the stages is submitted, and the FOV mask channel was not isolated.*
5. The integrated configuration reduces the distance between the training distribution and each external distribution in penultimate-layer feature space, on every external corpus examined, without any target-corpus statistic entering the transform (H-3). *In direction only — the magnitude of a distance reduction does not track the magnitude of any performance gain; the claim is mechanistic and carries no claim of diagnostic performance or device compatibility.*
6. A model trained on EyePACS with the pipeline transfers to APTOS 2019 without retraining, clearing the pre-registered generalisation ratio $G \geq 0.85$ and exceeding the baseline on every grade (H-4). *The baseline configuration also*

clears the threshold, so the criterion does not discriminate between the arms; the evidence lies in the comparison.

7. Model attention aligns more closely with expert pixel-level lesion annotation under the integrated configuration, on all four annotated lesion types and on both measures (H-5). *Alignment between model evidence and expert annotation, not clinical localisation of pathology; one annotated corpus and one fold; the qualitative examination on the clinical corpus was not carried out.*
8. Classification performance is maintained across every camera grouping examined and, substantively, the dispersion of performance across those groupings is reduced (H-6). *Both configurations clear the retention floor; two of the five groupings are the external clinical corpora themselves and furnish no independent replication; no part of this constitutes device certification.*
9. The integrated configuration attains higher absolute weighted F1 than the baseline on both external clinical corpora, each difference reaching the minimal clinically important difference with its confidence interval excluding zero, evaluated on each corpus separately (H-7). *Higher absolute external performance, not resistance to degradation — relative to their own in-domain levels the two configurations decline almost identically; the margin over the threshold on the second corpus is thin, and the lower bound of that interval lies below the minimal important difference.*
10. Three measures in common use for expressing external robustness — the degradation quantity, the generalisation ratio and the retention ratio — share a structural defect: each normalises or differences an arm's external performance against that same arm's in-domain performance, and therefore penalises a configuration for its in-domain strength. *Analytic and strictly descriptive: it rehabilitates no result of this work.*
11. A coupled thermal-optical model of fundus tissue response under laser exposure, and a modular architecture for an automated DR screening system in resource-limited environments. *Submitted as theoretical and design contributions respectively — the model is not clinically validated, and the architecture was neither prototyped nor field-tested.*

One further result is offered as an observation rather than as a statement for defence: in a preregistered evaluation on a corpus roughly one-seventieth the size of the training corpus, with both arms trained from the same initialisation, the integrated configuration was again the stronger, by a margin **comparable to, and not larger than**, the margin measured with abundant data. The clinical corpus involved cannot be redistributed, so that result is not externally reproducible.

The following are **not** submitted: clinical validity, readiness for unsupervised deployment, device certification, localisation of pathology, superiority over any named published system, and extension beyond the corpora, devices and hardware on which the results were obtained.

Theoretical significance

The work contributes a conceptual reframing of preprocessing as an integral model component that co-determines the feature space available to the CNN (paradigm P2), and places this reframing under direct empirical test against the end-to-end paradigm (P1). It provides a formal specification of the 8-stage pipeline, a mathematical formalisation of the dual-constraint CLAHE clip limit, the adaptive flat-field correction, and the ALO explainability metric, together with a rigorous statistical framework for validating preprocessing effects under class imbalance.

Practical significance

The integrated pipeline constitutes a transferable, interpretable, and device-robust preprocessing regime for automated DR diagnosis under constrained computational conditions. A modular automated DR screening system architecture is proposed, with preprocessing-engine, inference, telemedicine, and physician-in-the-loop decision-support modules, integrable with PACS/EHR systems and the national eHealth platform, and applicable to DR screening in rural and underserved regions of Kazakhstan. *(Implementation acts and approbation certificates: see appendices.)*

Reliability of the results

Reliability is ensured by: the use of large-scale and multiple public datasets (~35,126 EyePACS training images and seven additional datasets); 5-fold patient-level stratified cross-validation with strict leakage control; reporting of all primary metrics as mean ± standard deviation; a multi-metric assessment (weighted F1, ROC-AUC, Cohen's Kappa with quadratic weights, accuracy); statistical hypothesis testing (McNemar, DeLong), mixed-effects modelling across folds, bootstrap confidence intervals (≥ 1000 resamples), and Bonferroni/Holm correction; and success criteria fixed before the experiments that tested them — a criterion written once a result is known can always be satisfied, so the ordering is what makes the classification informative. The work is additionally placed against published systems without being ranked among them, since no two of the compared results share their task definition, corpus, reference standard and metric. Two limits of this apparatus are stated rather than reserved: correction for multiple comparisons is scoped to the single experiment within which the comparisons were planned, so no dissertation-wide error rate is claimed; and several evaluations rest on the models of one fitted fold, so their intervals quantify sampling of the evaluation corpus alone and understate total uncertainty in a known direction.

Approbation of the results and connection with scientific programmes

The results of the research were reported and discussed at international scientific forums, including the 3rd International Workshop on Digital Society (DS 2025, Istanbul, Türkiye, 28–30 October 2025), and published in international peer-reviewed journals.

The research direction corresponds to the state priorities of healthcare digitalisation and the development of artificial-intelligence technologies of the

Republic of Kazakhstan, in particular: the Concept for the Development of Artificial Intelligence for 2024–2029; the Address of the President of the Republic of Kazakhstan «Kazakhstan in the Era of Artificial Intelligence» (8 September 2025); and the Law of the Republic of Kazakhstan «On Artificial Intelligence» (No. 230-VIII, 17 November 2025). The work is carried out in accordance with subpara. 2, para. 3, art. 20 of the Law of the Republic of Kazakhstan «On Science».

Publications

The main results of the dissertation are published in 5 scientific works, including: articles in journals indexed by Scopus / Web of Science — 1 (*Eastern-European Journal of Enterprise Technologies*, Q3); a paper in international conference proceedings indexed by Scopus — 1 (*Procedia Computer Science*, DS 2025); and articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education (KKSON) of the Republic of Kazakhstan — 3 (*News of the NAS RK, Physico-Mathematical Series*; *Herald of the Kazakh-British Technical University*; *Herald of KazUTB*).

Articles in journals indexed by Scopus / Web of Science:

1. Sapakova S., Yesmukhamedov N., Sapakov A. Development of an image quality enhancement approach for diabetic retinopathy diagnosis // Eastern-European Journal of Enterprise Technologies. — 2025. — Vol. 4, No. 9(136). — P. 79–88. (Scopus, Q3). https://doi.org/10.15587/1729-4061.2025.335570

Papers in international conference proceedings indexed by Scopus:

2. Sapakova S., Yesmukhamedov N., Sapakov A., Yemberdiyeva A., Kozhamkulova Z. Methods for pre-processing and analysis of fundus images for detection of diabetic retinopathy // Procedia Computer Science. — 2025. — Vol. 272. — P. 496–501. (The 3rd International Workshop on Digital Society — DS 2025, Istanbul, Türkiye). https://doi.org/10.1016/j.procs.2025.10.237

Articles in journals recommended by the KKSON of the Republic of Kazakhstan:

3. Yesmukhamedov N.S., Sapakova S., Al-Haddad S.A.R., Daniyarova D. Development of an information system architecture for healthcare institutions using artificial intelligence // News of the National Academy of Sciences of the Republic of Kazakhstan, Physico-Mathematical Series. — 2025. — Vol. 2(354). — P. 74–91. https://doi.org/10.32014/2025.2518-1726.345
4. Yesmukhamedov N.S., Sapakova S.Z., Kozhamkulova Zh.Zh., Daniyarova D.R., Armankyzy R. Methods for preprocessing and analysis of fundus images for diabetic retinopathy detection // Herald of the Kazakh-British Technical University. — 2025. — No. 4(75). — Vol. 22. — P. 119–130. https://doi.org/10.55452/1998-6688-2025-22-4-119-130
5. Sapakova S.Z., Daniyarova D.R., Yesmukhamedov N.S., Armankyzy R., Yemberdiyeva A.B., Kaldybaeva A.S. Mathematical modeling of laser exposure on fundus tissues in the treatment of diabetic retinopathy // Herald of KazUTB. — 2024. — Vol. 2, No. 27-740. https://doi.org/10.58805/kazutb.v.2.27-740

Main content of the work

The introduction substantiates the relevance of the topic, formulates the aim, objectives, object and subject of the research, the central hypothesis, the scientific novelty, the statements submitted for defense, the methodological basis, the theoretical and practical significance, the reliability of the results, the empirical basis, the approbation, the connection with scientific programmes, and the publications, and describes the structure and length of the dissertation.

Chapter 1 — "Problem Domain Analysis and Current State of Automated Diabetic Retinopathy Diagnosis" examines the medical and epidemiological context of DR and its clinical grading, the screening requirements of resource-limited settings, the sources of fundus image quality degradation and device-specific variability, deep-learning approaches to retinal image classification (CNN architectures, transfer and self-supervised pretraining, explainability), and provides a critical analysis of existing automated DR screening systems, concluding with the formulation of the research problem and the justification of the research direction.

Chapter 2 — "Theoretical Foundations of Image Preprocessing and Deep Learning for Fundus Image Analysis" establishes the mathematical foundations of image enhancement (histogram equalisation, dual-constraint CLAHE, spatial filtering), the theoretical framework of CNNs (convolution, pooling, loss functions for imbalanced data, regularisation), transfer and in-domain self-supervised representation learning, the mathematical modelling of laser-tissue interaction in retinal therapy, explainability (CAM, Grad-CAM, ALO, IoU), and image quality metrics for preprocessing evaluation.

Chapter 3 — "Methodology of Integrated Preprocessing-CNN Pipeline Design" formalises the unified 8-stage preprocessing pipeline and the modified dual-constraint CLAHE algorithm, the augmentation strategy, and the external image ingestion protocol; specifies the CNN architectures (ResNet-50 and EfficientNet-B3); details the transfer-learning and ophthalmology-specific self-supervised pretraining methodology, the architecture adaptation for five-class classification, and the weighted (Focal) loss formulation; and defines the multi-metric evaluation and statistical reliability framework.

Chapter 4 — "Experimental Research — Preprocessing Impact on CNN Diagnostic Performance" presents the tiered dataset architecture and eight investigations: Experiment 1 (integrated-pipeline dominance via a restored 2×2 factorial, configs A–D), Experiment 2 (cumulative stage ablation, CLAHE threshold and flat-field σ sweeps with image-quality metrics), the direct measurement of source-to-target domain distance across six external corpora (H-3), Experiment 3 (zero-shot cross-dataset transfer to APTOS 2019), Experiment 4 (Grad-CAM explainability with quantitative ALO/IoU on IDRiD), Experiment 5 (external clinical performance on IDRiD and Messidor-2), Experiment 6 (device domain shift on DDR, ODIR-5K, RFMiD), and Experiment 7 (small-data training on IDRiD with the Clinical dataset held out).

Chapter 5 — "Reliability Validation and Comparative Analysis" consolidates the explainability results, the statistical validation (bootstrap confidence intervals and mixed-effects modelling, final claim-strength classifications), the placement of the work against published systems without ranking it among them, and the performance–complexity trade-off, and states the limitations and boundary conditions of the proposed approach.

Chapter 6 — "Architecture of an Automated DR Screening System for Resource-Limited Environments" specifies the functional and non-functional requirements, the modular architecture with PACS and EHR integration, the AI processing module (configurable preprocessing engine and inference module), clinical workflow integration (telemedicine, portable-device and national eHealth-platform support, physician-in-the-loop decision support), and the data security and regulatory-compliance framework (GDPR/HIPAA-aligned), with its applicability to Kazakhstan healthcare infrastructure.

Author's personal contribution

The principal results of the dissertation were obtained by the author personally: the conceptual reframing of preprocessing as an integral component of the diagnostic model (paradigm P2); the design and formal specification of the 8-stage preprocessing pipeline, including the dual-constraint stochastic CLAHE, the adaptive flat-field correction, and the FOV-mask input channel; the Attention–Lesion Overlap (ALO) explainability metric; the controlled experimental design and statistical validation framework; and the architecture of the automated DR screening system. In the joint publications the author developed the fundus image preprocessing and analysis methods and prepared the corresponding parts of the manuscripts.

Structure and length of the dissertation

The dissertation consists of an introduction, six chapters, a conclusion, a list of references used, and appendices. The work is supplemented with tables, figures, and the source code of the preprocessing pipeline.